IDENTIFICATION IN MODELS FOR MATCHED PANEL DATA WITH TWO-SIDED RANDOM EFFECTS

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Abstract

This paper is concerned with models for matched worker-firm data in the presence of both worker and firm heterogeneity. We show that models with complementarity and sorting can be nonparametrically identified from short panel data while treating both worker and firm heterogeneity as discrete random effects. This paradigm is different from the framework of Bonhomme, Lamadon and Manresa (2019a), where identification results are derived under the assumption that worker effects are random but firm heterogeneity is observed. Our identification approach is constructive and our results appear to be the first of their kind in the context of matched panel data problems.

JEL Classification: C23, J31, J62

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Introduction

Matched panel data are often used to study the interaction between two types of units over a period of time. The importance of unobserved heterogeneity across the units in such data is well recognized and understanding its implications has received considerable attention.

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The seminal work of Abowd, Kramarz and Margolis (1999), for example, was concerned with matched worker-firm data. They regressed wages on worker and firm fixed effects to quantify the degree of heterogeneity in wages coming from, respectively, worker and firm heterogeneity and used the fixed-effect decomposition to investigate sorting patterns between workers and firms. Such decompositions have become a workhorse tool in the labor literature and have been widely adopted in other fields; teacher value-added studies (McCaffrey, Lockwood, Koretz, Louis, and Hamilton 2004, Chetty, Friedman and Rockoff 2014) are one example. To maintain focus, for the remainder of this paper we will use terminology from the worker-firm application, even though everything to follow is not specific to that setting.

The regression approach of Abowd, Kramarz and Margolis (1999) has come under increased scrutiny. The linearity of the model does not permit any form of complementarity between workers and firms, which is at odds with theoretical models (see, e.g., Shimer and Smith 2000 or Eeckhout and Kircher 2011). It is also inherently static, as it rules out any dynamics in wages, even within job spells (see Lentz, Piyapromdee and Robin 2023, pp. 2416-2417, for a discussion). Furthermore, their fixed-effect decompositions produce highly unreliable results in the type of data to which they are usually applied (Jochmans and Weidner 2019).[1]

In influential work Bonhomme, Lamadon and Manresa (2019a) proposed an alternative framework that permits both complementarity and sorting. Their approach is based on the assumption that worker and firm heterogeneity is discrete and hinges on the presumption that firms can be consistently clustered by type from the cross-sectional distribution of

[1]The presence of bias in estimators of variance components is now well understood. It can be corrected for using any of a variety of estimators (see Andrews, Gill, Schank and Upward 2008, Kline, Saggio and Sølvsten 2020, Azkarate-Askasua and Zerecero 2024, and Babet, Godechot and Palladino 2025). However, even after bias correction, the estimators exhibit highly non-standard asymptotic behavior that, to date, is not fully understood (see Kline, Saggio and Sølvsten 2020 for results in problems with a particular connectivity structure). The bias in the empirical distribution of the estimated worker and firm fixed effects, or in any of its functionals other than second moments, cannot easily be corrected; see Jochmans and Weidner (2019) for a discussion.

wages in an initial step. Given such a consistent classification the firm types can be treated as observed in the data, so that the model effectively features only one-way heterogeneity (in the form of the worker effect). Arguments reminiscent of those used in the literature on multivariate finite mixtures (see Schennach 2020, for example, for a recent overview of this literature) can then be used to identify certain aspects of the model from short panel data.

The theory underlying the consistent clustering of firms requires minimal firm size to diverge with the sample size. Such a paradigm does not align well with most available employer-employee data sets, where many small firms are present (see He and Robin 2025 for a similar discussion). A framework that allows for small firms to exist in large samples would be one where the number of workers and firms grow at the same rate. In such a case, average firm size is bounded, and so neither worker nor firm effects can be estimated consistently. Such a setting thus requires treating both the worker and firm heterogeneity as random effects. This is not compatible with Bonhomme, Lamadon and Manresa (2019a). Furthermore, to date, it is not known under what type of conditions models with such two-sided unobserved heterogeneity can be identified, if at all.[2] This paper aims to make progress on this question.

We provide conditions for identification of all primitive parameters in a stationary version of the model of Bonhomme, Lamadon and Manresa (2019a).[3] The primitives are

[2]Such random-effect models have nonetheless been estimated; see, for example, Beffy, Kamionka, Kramarz and Robert (2003), Bonhomme, Lamadon and Manresa (2019b), and Abowd, McKinney and Schmutte (2019).

[3]This is in line with the empirical practice of estimating stationary specifications. While the theoretical setup in Bonhomme, Lamadon and Manresa (2019a) permits non-stationarity, their baseline empirical specification is stationary. Furthermore, their results show that a non-stationarity version gives very similar results to their baseline model. Lamadon, Lise, Meghir and Robin (2025, pp. 25, Footnote 7) also report empirical support for their stationarity assumption. From a theoretical perspective, working under stationarity allows us to recover all model parameters, which contrasts with Bonhomme, Lamadon and Manresa (2019a, Theorems 1 and 2). In dynamic models with latent variables lack of stationarity usually prevents full identification; compare, for example, the results in Higgins and Jochmans (2023) to those in Hu and Shum (2012).

the distributions of wages and mobility decisions conditional on the worker and firm effects, which are informative about heterogeneity and complementarity, and the joint distribution of worker and firm effects, which is informative about sorting. Our setup is sufficiently rich to cover models à la Shimer and Smith (2000) and Hagedorn, Law and Manovskii (2017), where workers pass through unemployment before taking up a new job, while also allowing for on-the-job search, which occurs in wage-ladder models such as Burdett and Mortensen (1998) or Delacroix and Shi (2006).

The identification argument in Bonhomme, Lamadon and Manresa (2019a) relies on identification techniques for one-sided random-effect models. Such techniques are built around conditional-independence assumptions that are not available when heterogeneity is two-sided. We show that certain (different) conditional-independence conditions are nonetheless present that can be used to achieve identification. On the firm side, we exploit that the worker types are independent within a given firm. On the worker side, we rely on identifying restrictions coming from the assumption that employment history becomes irrelevant when a worker goes through unemployment, as in Shimer and Smith (2000), for example. This is important, as observing only job-to-job transitions does not suffice to create sufficient identifying content in the two-sided case. This is so because on-the-job search creates time-series dependence in the random effect of the firms at which a given worker is employed, conditional on the worker effect. Because in Bonhomme, Lamadon and Manresa (2019a) the firm effects are known, the trajectory of workers across the different firm types is directly observable, and so any such dependence does not create complications for identification.

Identification of the full model is achieved from three-wave panel data under rather intuitive conditions, in line with the literature on latent-variable models and reminiscent of some of the requirements in Bonhomme, Lamadon and Manresa (2019a) and Lentz, Piyapromdee and Robin (2023). Importantly, however, these conditions demand neither sorting between workers and firm to be present nor the existence of complementarity in the wage equation. In the absence of on-the-job search, identification is achieved under weaker conditions. Our identification proof is constructive in that, in principle, it points to the construction of a nonparametric estimator. However, the chief goal of the analysis here is to consider the type of variation in the data that is needed to be able to credibly identify models for employer-employee data with two-sided unobserved heterogeneity. In practice, as is done in the empirical literature, one may prefer to proceed with a more parametric approach.

In Section 1 we formally set up the model under study and define the parameters of interest. In Section 2 we provide the assumptions under which we will work and connect them to the literature before stating our main identification result. In Section 3 we provide the associated proof. The different steps of the argument highlight the identifying variation that we rely on and show how and where the different assumptions come into play. As such, it is instructive; this section can nevertheless be skipped if desired. We end the paper with some concluding remarks.

1 The model

We consider panel data on workers followed over time. For each worker i and each time period t we observe the worker's wage, w_{it}, together with an indicator, x_{it}, concerning job mobility between periods t and $t+1$. Specifically, $x_{it} = e$ if the worker stays with the same firm, $x_{it} = s$ if the worker separates from his current employer before matching with a new firm in the subsequent period, and $x_{it} = c$ if the worker changes employer between periods as a consequence of a job-to-job transition. The different cases are important to cover both models with and without on-the-job search. We also know the identity of the firm where worker i was employed at time t, say $f(i,t)$. As Lentz, Piyapromdee and Robin (2023) and Lamadon, Lise, Meghir and Robin (2025) we consider stationary data, that is, we will look at an economy in steady state.

Worker i and firm f are characterized by unobserved heterogeneity ϕ_i and ψ_f. We will follow the literature (e.g., Bonhomme, Lamadon and Manresa 2019a, pp. 703 and 708) and presume that both types of heterogeneity are discrete, with a known number of support points.[4,5] Let ψ_{it} be shorthand notation for $\psi_{f(i,t)}$, that is, the effect of the firm where worker i is employed at time t. The joint distribution of the worker and firm heterogeneity then is

$$p(\phi, \psi) := \mathbb{P}(\phi_i = \phi, \psi_{it} = \psi).$$

The wage and mobility processes are initialized in the following manner. First, workers independently draw their type with probability $p(\phi) := \mathbb{P}(\phi_i = \phi) = \int p(\phi, \psi)\, d\psi$. Firms, in turn, draw their type independently according to $p(\psi) := \mathbb{P}(\psi_f = \psi) = \int p(\phi, \psi)\, d\phi$. An initial allocation then follows from assigning a worker of type ϕ to a firm of type ψ with probability

$$p_\phi(\psi) := \frac{p(\phi, \psi)}{p(\phi)}.$$

First period wages w_{i1} are drawn from the conditional distribution $Q_{\phi_i, \psi_{i1}}$, where we write

$$Q_{\phi,\psi}(w) := \mathbb{P}(w_{it} \leq w | \phi_i = \phi, \psi_{it} = \psi),$$

independently for each worker. Next, the match quality between the worker and his current firm is evaluated with a draw x_{i1} from $r_{\phi_i, \psi_{i1}}$, where

$$r_{\phi,\psi}(x) := \mathbb{P}(x_{it} = x | \phi_i = \phi, \psi_{it} = \psi), \qquad x \in \{s, c, e\}.$$

There are then three possibilities.

First consider the case where $x_{i1} = e$. Then the worker remains in the same firm, so that $f(i, 2) = f(i, 1)$ and, therefore, $\psi_{i2} = \psi_{i1}$. Within job spells wages are allowed

[4]Under the assumptions spelled out below the smallest number of support points for the distribution of the worker effect and for the firm effect that can rationalize our model are in fact identified as the rank of the bivariate distribution of (w_{i1}, w_{i2}) for workers that change employment between periods one and two as a consequence of separation, and as the rank of the bivariate distribution of $(w_{i_1 1}, w_{i_2 1})$ for pairs of distinct workers (i_1, i_2) employed in the same firm in the first time period. This result is in line with Kasahara and Shimotsu (2014), Bonhomme, Jochmans and Robin (2016b), and Kwon and Mbakop (2021). While seemingly new in the current context, we do not focus on it here.

[5]We will nevertheless use integral notation rather than sums when marginalizing with respect to a discrete measure as this lightens some of the expressions in the sequel.

to exhibit Markovian dependence conditional on both the worker and firm effect. The transition kernel is

$$Q_{\phi,\psi,w}(w') \coloneqq \mathbb{P}(w_{it} \leq w' | w_{it-1} = w, x_{it-1} = e, \phi_i = \phi, \psi_{it} = \psi),$$

whose steady-state distribution is $Q_{\phi,\psi}$. Thus, when $x_{i1} = e$, the second-period wage w_{i2} is a draw from $Q_{\phi_i,\psi_{i2},w_{i1}}$.

For workers that change employer between periods there are two cases to consider. If $x_{i1} = s$ employment at firm $f(i,1)$ is terminated. The worker passes through unemployment and draws a new firm type ψ_{i2} from the conditional distribution p_{ϕ_i}. If $x_{i1} = c$, the worker changes employer as a consequence of on-the-job search, without passing through unemployment. In this case, the new firm type is a draw from $k_{\phi_i,\psi_{i1}}$, for transition kernel

$$k_{\phi,\psi}(\psi') \coloneqq \mathbb{P}(\psi_{it} = \psi' | \phi_i = \phi, \psi_{it-1} = \psi),$$

whose steady-state distribution is p_ϕ. In either of the two cases, a new wage is drawn from the implied $Q_{\phi_i,\psi_{i2}}$.

For each of the three possible cases, once wages realize the match quality between the worker and his current employer is again evaluated and job mobility decisions are made according to $r_{\phi_i,\psi_{i2}}$. The process then repeats itself for each remaining period in the panel.

2 Identification

Our aim is to nonparametrically identify the steady-state distributions $Q_{\phi,\psi}$, the transition kernels $Q_{\phi,\psi,w}$, the mobility distributions $r_{\phi,\psi}$, as well as the joint distribution of worker and firm types $p(\phi,\psi)$ and the transition kernels $k_{\phi,\psi}$. Of course, because the types are latent, it is understood that identification here will be up to an arbitrary relabelling of the types. In empirical applications one often given empirical content to the types (such as them being a measure of ability or efficiency, for example). In such cases, if desired, types can be ordered by a functional (such as the mean) of the wage distributions conditional on only worker or firm type, that is,

$$Q_\phi(w) \coloneqq \int Q_{\phi,\psi}(w)\, p_\phi(\psi)\, d\psi, \qquad \text{and} \qquad Q_\psi(w) \coloneqq \int Q_{\phi,\psi}(w)\, p_\psi(\phi)\, d\phi,$$

where we let $p_\psi(\phi) := p(\phi,\psi)/p(\psi)$ in analogy to $p_\phi(\psi)$. For our purposes, however, such an ordering is not needed and, hence, is irrelevant.

Our main result requires three assumptions. To state the first of these assumptions, let

$$P_\phi(w) := \mathbb{P}(w_{it} \leq w, x_{it} = s | \phi_i = \phi) = \int r_{\phi,\psi}(s)\, Q_{\phi,\psi}(w)\, p_\phi(\psi)\, d\psi.$$

Assumption 1. *The distributions P_ϕ and Q_ϕ, and Q_ψ are linearly independent in ϕ and ψ, respectively.*

Assumption 1 demands that changes in ϕ and ψ affect wages, which is intuitive. A simple sufficient condition is that the distributions $Q_{\phi,\psi}$ are linearly independent in (ϕ,ψ), but this is much stronger than needed. Assumption 1 permits, but does not require, the presence of complementarity; it can accommodate linear wage processes with additively-separable worker and firm heterogeneity, such as the model of Abowd, Kramarz and Margolis (1999). Conditions of this type are typical in the analysis of multivariate latent-variable models. In Bonhomme, Lamadon and Manresa (2019a) the corresponding restriction on the wage distribution conditional on the worker effect appears in Assumption 3(ii), where the $Q_{\phi,\psi}$ are required to be linearly independent in ϕ for each ψ. On the firm side, their Assumption B1, in turn, demands the conditional wage distributions Q_ψ to be well separated across the different ψ.

The second assumption is a support condition.

Assumption 2. *For all (ϕ,ψ) it holds that (i) $0 < p(\phi,\psi) < 1$ and that (ii) $r_{\phi,\psi}(s) > 0$.*

Assumption 2 has two parts. Part (i) is a full-support condition on the distribution of the latent types. It states that all matches between worker and firm types are possible. It does not state that a transition between any two firm types is possible for all worker types, which would be restrictive. Such moves involve the transition kernel $k_{\phi,\psi}$, which we leave entirely unrestricted. It is useful to note that Part (i) is implicit in Lentz, Piyapromdee and Robin (2023, pp. 2443). They impose that

$$p_{\psi,\psi'}(\phi) := \mathbb{P}(\phi_i = \phi | \psi_{i1} = \psi, \psi_{i2} = \psi', x_{i1} \neq e) \propto p(\phi,\psi)\,\{r_{\phi,\psi}(s)\, p_\phi(\psi') + r_{\phi,\psi}(c)\, k_{\phi,\psi}(\psi')\}$$

is strictly positive for all ϕ and (ψ, ψ'). This clearly implies Part (i). Bonhomme, Lamadon and Manresa (2019a) use a slightly more elaborate argument based on the existence of certain types of cycles in the graph of firm types.[6] In either case, they further need these probabilities to vary sufficiently, in a specific sense; see Bonhomme, Lamadon and Manresa (2019a, pp. 708–709). Among other things, this requires that $p_{\psi,\psi'}$ depends on both ψ and ψ' and that $p_{\psi,\psi'} \neq p_{\psi',\psi}$. Such asymmetry is crucial to their approach. It rules out, for example, the model of Shimer and Smith (2000), where $r_{\phi,\psi}(c) = 0$ and $r_{\phi,\psi}(s)$ does not depend on ψ (or, indeed, on ϕ). Part (ii), in turn, simply states that any match between a worker and firm can terminate.

Assumptions 1 and 2 suffice to identify all parameters in a model that does not feature on-the-job search, that is, if $r_{\phi,\psi}(c) = 0$ for all (ϕ, ψ). To obtain full identification in the complete model we impose an additional assumption.

Assumption 3. *The distributions $Q_{\phi,\psi}$ are linearly independent in ψ for each ϕ.*

This requirement is the mirror image of Assumption 3(ii) in Bonhomme, Lamadon and Manresa (2019a).

The following theorem states our main result.

Theorem 1. *Let Assumptions 1–3 hold. Then*

(i) the wage distributions $Q_{\phi,\psi}$ and transition kernels $Q_{\phi,\psi,w}$,

(ii) the mobility distributions $r_{\phi,\psi}$ and transition kernels $k_{\phi,\psi}$, and

(iii) the type distribution $p(\phi, \psi)$

[6] A connecting cycle consists of sequences of firm types $\psi_1, \ldots, \psi_m$ and $\psi'_1, \ldots, \psi'_m$ for which (i) $p_{\psi_1,\psi'_1}(\phi)\, p_{\psi_2,\psi'_2}(\phi) \cdots p_{\psi_m,\psi'_m}(\phi) > 0$ and (ii) $p_{\psi_2,\psi'_1}(\phi)\, p_{\psi_3,\psi'_2}(\phi) \cdots p_{\psi_1,\psi'_m}(\phi) > 0$ for all ϕ. Bonhomme, Lamadon and Manresa (2019a) require that, for each (ψ, ψ'), such cycles exist for (a) $\psi_1 = \psi$ and $\psi_r = \psi'$ for some $1 \leq r \leq m$ and for (b) $\psi_1 = \psi$ and $\psi'_r = \psi'$ for some $1 \leq r \leq m$. In the simplest example, where there are two firm types, this effectively requires that $p(\phi, \psi) > 0$ for all (ϕ, ψ). For Case (a), they further require the ratio of (i) to (ii) to be different for each ϕ. This allows for identification through a spectral decomposition.

are all nonparametrically identified up to relabelling of (ϕ, ψ) from panel data on wage trajectories and mobility patterns spanning three time periods.

For the interested reader, the proof of the theorem is given in the next section.

3 Proof of Theorem 1

The proof of our main result is lengthy but instructive. We proceed in six steps. The first two of these serve to identify auxiliary parameters that will be used to identify the model parameters in the remaining steps.

The first step is concerned with identifying the distribution of wages conditional on the worker type alone, that is, the functions Q_ϕ, up to an arbitrary ordering of the ϕ. To do so we use the panel dimension of our setup. More precisely, we exploit the observation that in our model wages across job spells are independent conditional on the worker effect when passing through unemployment. To see how this is helpful for our purposes consider the joint probability

$$\mathbb{P}(w_{i1} \leq w_1, x_{i1} = s, w_{i2} \leq w_2, x_{i2} = s, w_{i3} \leq w_3) \tag{3.1}$$

for chosen values (w_1, w_2, w_3). Here, workers switch employer between the first and second period, and again between the second and third period. The probability of this happening is non-zero under Assumption 2. By conditional independence, the joint probability in (3.1) factors as

$$\int P_\phi(w_1)\, P_\phi(w_2)\, Q_\phi(w_3)\, p(\phi)\, d\phi,$$

which is a tri-variate finite mixture. By Assumption 1 the distributions Q_ϕ and P_ϕ, seen as functions of ϕ, are linearly independent. By Assumption 2, $p(\phi) > 0$ for all ϕ. From Allman, Matias and Rhodes (2009, Theorem 8) or Bonhomme, Jochmans and Robin (2016a, Theorem 2), for example, it then follows that the mixture distributions, P_ϕ and Q_ϕ, and the mixing weights $p(\phi)$ are nonparametrically identified for all ϕ up to label swapping. A constructive approach to recovering these distributions is in Bonhomme, Jochmans and Robin (2016a).

The second step of our proof, in turn, identifies the distribution of wages conditional on the firm type alone, that is, the functions Q_ψ, again up to an arbitrary ordering of the ψ. To do this we exploit the cross-sectional dimension of our problem and the fact that firm identities are known. Consider the cross-sectional distribution of wages of distinct workers (i_1, i_2, i_3) employed by the same firm f in time period t. It is helpful to make the dependence of wages on the firm explicit, by writing w_{ift} for the wage of worker i earned in firm f at time t. Then we can write the probability distribution in question, evaluated at (w_1, w_2, w_3), as

$$\mathbb{P}(w_{i_1ft} \leq w_1, w_{i_2ft} \leq w_2, w_{i_3ft} \leq w_3). \tag{3.2}$$

Wages of different workers employed at the same firm are independent conditional on the firm effect. Therefore, their joint probability in (3.2) factors as

$$\int Q_\psi(w_1)\, Q_\psi(w_2)\, Q_\psi(w_3)\, p(\psi)\, d\psi,$$

which is again a tri-variate finite mixture. In the same way as before, this representation implies that the Q_ψ are nonparametrically identified up to a relabelling of the firm types ψ.

In the third step of our proof we use the results obtained so far to recover the conditional wage distributions $Q_{\phi,\psi}$ for the labelling of worker and firm types from the previous two steps. This is done by looking at the joint distribution of wages for two distinct workers initially employed at the same firm, together with the next period's wage of one of them that separates from the employer at the end of the period. The joint distribution in question, as a function of (w_1, w, w_2) is

$$\mathbb{P}(w_{i_1f1} \leq w_1, w_{i_2f1} \leq w, x_{i_21} = s, w_{i_22} \leq w_2).$$

Under the dynamics of our model this probability can be written in terms of the model primitives as

$$\int\!\!\int Q_\phi(w_2)\, H(w, \phi, \psi)\, Q_\psi(w_1)\, d\phi\, d\psi, \tag{3.3}$$

where

$$H(w, \phi, \psi) \coloneqq \mathbb{P}(w_{it} \leq w, x_{it} = s, \phi_i = \phi, \psi_{it} = \psi) = Q_{\phi,\psi}(w)\, r_{\phi,\psi}(s)\, p(\phi, \psi).$$

The latter joint probability is identified because the Q_ϕ and Q_ψ are identified and are linearly independent. To see this take a collection of values for the wage, $w_1, \ldots, w_m$ for some finite integer m so that the matrices $(A)_{v,\phi} \coloneqq Q_\phi(w_v)$ and $(B)_{v,\psi} \coloneqq Q_\psi(w_v)$ have maximal column rank. By Assumption 1 such a set of values exists. Further, for any w, let $(C_w)_{v_1,v_2} \coloneqq \mathbb{P}(w_{i_1f1} \leq w_{v_1}, w_{i_2f1} \leq w, x_{i_21} = s, w_{i_22} \leq w_{v_2})$ and $(D_w)_{\phi,\psi} \coloneqq H(w,\phi,\psi)$. Then, from (3.3), $C_w = A\, D_w B'$ so that $D_w = (A'A)^{-1}A'C_wB(B'B)^{-1}$, which contains the $H(w,\phi,\psi)$ for any w, is identified. A value w of particular interest is $w = +\infty$, for which

$$h(\phi,\psi) \coloneqq H(+\infty,\phi,\psi) = \mathbb{P}(x_{it} = s, \phi_i = \phi, \psi_{it} = \psi) = r_{\phi,\psi}(s)\, p(\phi,\psi).$$

Under Assumption 2 the probability $h(\phi,\psi)$ is strictly positive. Therefore, for any chosen value w we have that

$$Q_{\phi,\psi}(w) = \frac{H(w,\phi,\psi)}{h(\phi,\psi)}$$

is nonparametrically identified for each (ϕ,ψ) and all w, and this up to the same labelling of worker and firm types as before.

In the fourth step of our proof we follow a similar approach as in the previous step to identify the transition kernel of the wage, $Q_{\phi,\psi,w}$. Now, rather than looking at workers who switch employer after the first period, we look at workers that separate in the second period. The relevant probability distribution, now seen as a function of (w_1, w, w', w_2) is equal to

$$\mathbb{P}(w_{i_1f1} \leq w_1, w_{i_2f1} \leq w, x_{i_21} = e, w_{i_22} \leq w', x_{i_22} = s, w_{i_23} \leq w_2).$$

It factors as

$$\iint Q_\phi(w_2)\, G(w,w',\phi,\psi)\, Q_\psi(w_1)\, d\phi\, d\psi, \tag{3.4}$$

where

$$G(w,w',\phi,\psi) \coloneqq \mathbb{P}(w_{it-1} \leq w, w_{it} \leq w', x_{it-1} = e, x_{it} = s, \phi_i = \phi, \psi_{it} = \psi).$$

Observe that $G(w,w',\phi,\psi) = Q_{\phi,\psi}(w,w')\, r_{\phi,\psi}(s)\, r_{\phi,\psi}(e)\, p(\phi,\psi)$, where we have used the shorthand notation

$$Q_{\phi,\psi}(w,w') \coloneqq \mathbb{P}(w_{it} \leq w, w_{it+1} \leq w' | x_{it} = e, \phi_i = \phi, \psi_{it} = \psi)$$

for the joint distribution of two wage observations within a given employment spell. From this decomposition, by the same argument as used for the function H before, the function G is identified up to the same labelling of worker and firm types. From this we may then recover

$$g(\phi, \psi) \coloneqq \mathbb{P}(x_{it-1} = e, x_{it} = s, \phi_i = \phi, \psi_{it} = \psi) = r_{\phi,\psi}(e)\, r_{\phi,\psi}(s)\, p(\phi, \psi),$$

as $G(+\infty, +\infty, \phi, \psi)$, given which we find

$$Q_{\phi,\psi}(w, w') = \frac{G(w, w', \phi, \psi)}{g(\phi, \psi)}$$

for any pair (w, w') and all (ϕ, ψ). From this we then equally obtain the conditional distribution $Q_{\phi,\psi,w}$.

In the fifth step of our proof we recover the conditional distribution of mobility decisions, $r_{\phi,\psi}$, and the joint distribution $p(\phi, \psi)$ of worker and firm types. First note that, from above,

$$r_{\phi,\psi}(e) = \frac{g(\phi, \psi)}{h(\phi, \psi)}.$$

In a setting without on-the-job search, where $r_{\phi,\psi}(c) = 0$, we then immediately recover $r_{\phi,\psi}(s) = 1 - r_{\phi,\psi}(e)$ and with it,

$$p(\phi, \psi) = \frac{h(\phi, \psi)^2}{h(\phi, \psi) - g(\phi, \psi)},$$

all again up to the same ordering of worker and firm types. Therefore, in the absence of on-the-job search, all parameters of the model have been shown to be identified and the proof ends here. This case, therefore, only uses three time periods of data and does not require Assumption 3.

In the presence of job-to-job transitions, the above gives $r_{\phi,\psi}(e)$, but we need a more elaborate argument for the remaining parameters. Recall that we have previously identified $p(\phi)$ in the first step. Then

$$\frac{h(\phi, \psi)}{p(\phi)} = r_{\phi,\psi}(s)\, p_\phi(\psi).$$

From this we can learn $r_{\phi,\psi}(s)$ and, consequently, also $r_{\phi,\psi}(c) = 1 - r_{\phi,\psi}(e) - r_{\phi,\psi}(s)$ given knowledge of p_ϕ. Under Assumption 3 the latter is identified from a least-squares projection of $Q_\phi(w) = \int Q_{\phi,\psi}(w)\, p_\phi(\psi)\, d\psi$ on $Q_{\phi,\psi}(w)$ for each ϕ. Again taking a set of wages values $w_1, \ldots, w_m$ for some m, we can construct, for each ϕ, the vector $(a_\phi)_v := Q_\phi(w_v)$ and matrix $(B_\phi)_{v,\psi} := Q_{\phi,\psi}(w_v)$, which have both been shown to be identified. For each ϕ there exists a set of such values for which B_ϕ has full column rank, and so $p_\phi = (B'_\phi B_\phi)^{-1} B'_\phi a_\phi$ follows. Then, clearly, $p(\phi, \psi) = p_\phi(\psi)\, p(\phi)$ is also identified given that each of the distributions on the right-hand side are now known.

In our sixth step it only remains to identify the transition kernels $k_{\phi,\psi}$ of the firm types for all worker types ϕ. To see how these can be identified, consider

$$M_\phi(w_1, w_2) := \mathbb{P}(w_{i1} \le w_1, x_{i1} = c, w_{i2} \le w_2, x_{i2} = s | \phi_i = \phi).$$

Clearly,

$$M_\phi(w_1, w_2) = \iint r_{\phi,\psi'}(s)\, Q_{\phi,\psi'}(w_2)\, k_{\phi,\psi}(\psi')\, r_{\phi,\psi}(c)\, Q_{\phi,\psi}(w_1)\, p_\phi(\psi)\, d\psi'\, d\psi,$$

where of all the quantities under the double integral, only $k_{\phi,\psi}$ has not yet been shown to be identified. By Assumption 3, it can be recovered if the conditional distribution M_ϕ on the left-hand side can be treated as known. If so, letting $(K_\phi)_{\psi',\psi} := k_{\phi,\psi}(\psi')$, and constructing the diagonal matrices $(D_{\phi,c})_{\psi,\psi} := r_{\phi,\psi}(c)$, $(D_{\phi,s})_{\psi,\psi} := r_{\phi,\psi}(s)$, and $(D_\phi)_{\psi,\psi} := p_\phi(\psi)$, and the matrices $(A_\phi)_{v,\psi} := Q_{\phi,\psi}(w_v)$ and $(C_\phi)_{v_1,v_2} := M_\phi(w_{v_1}, w_{v_2})$ for a grid of values $w_1, \ldots, w_m$ for which Assumption 3 holds, we have $C_\phi = A_\phi D_{\phi,s} K_\phi D_{\phi,c} D_\phi A'_\phi$ and, on inverting the system, that

$$K_\phi\, D_{\phi,c} = D_{\phi,s}^{-1} (A'_\phi A_\phi)^{-1} A'_\phi C_\phi A_\phi (A'_\phi A_\phi)^{-1} D_\phi^{-1}$$

is identified for each ϕ. Thus, the probability of a worker of type ϕ transitioning from a firm of type ψ to a firm of type ψ' is identified for any worker-firm pair (ϕ, ψ) for which a job-to-job transition can occur, and so the conditioning event in the conditional probability distribution $k_{\phi,\psi}$ is well defined. It remains only to show that the M_ϕ are identified. To do so we use that

$$\mathbb{P}(w_{i1} \le w_1, x_{i1} = c, w_{i2} \le w_2, x_{i2} = s, w_{i3} \le w_3)$$

factors as

$$\int M_\phi(w_1, w_2)\, Q_\phi(w_3)\, p(\phi)\, d\phi.$$

Here, all distributions under the integral sign except for M_ϕ are known. Furthermore, the Q_ϕ are linearly independent in ϕ and $p(\phi) > 0$ for all ϕ. Hence, we can solve for M_ϕ using a least-squares approach. For any chosen pair of values (w, w') and a collection of values $w_1, \ldots, w_m$ for which the matrix $(A)_{v,\phi} \coloneqq Q_\phi(w_v)$ has full column rank, let $(a)_v \coloneqq \mathbb{P}(w_{i1} \leq w, x_{i1} = c, w_{i2} \leq w', x_{i2} = s, w_{i3} \leq w_v)$ and redefine $(D)_{\phi,\phi} \coloneqq p(\phi)$. Then $a = ADc$ for $(c)_\phi \coloneqq M_\phi(w, w')$, and so

$$c = D^{-1}(A'A)^{-1}A'a$$

With all parameters identified for the general case, the proof of the theorem is complete. □

Conclusion

We have given an identification result for a model for matched employer-employee data with discrete two-sided unobserved heterogeneity. Our setup crucially differs from the one followed in Bonhomme, Lamadon and Manresa (2019a), Lentz, Piyapromdee and Robin (2023), and others in that we treat the heterogeneity on both sides as random effects. Doing so by-passes the need to consistently estimate the heterogeneity on (at least) one side. The latter is fundamental to the identification results available to date but is difficult to maintain in typical data sets. Moreover, our arguments are consistent with the presence of small firms. We flesh-out the kind of variation that is needed to nonparametrically recover the primitive parameters in a model that allows for complementarity, sorting, and on-the-job search. Under certain rank conditions, identification is achieved from three-wave panel data.

Variations of our model can be entertained, and there appears to be scope for further generalization. For example, an older version of this paper contained a result for a model with endogenous mobility (see, e.g., Abowd, McKinney and Schmutte 2019), where mobility decisions depend on current wages even after conditioning on worker and firm heterogeneity.

References

Abowd, J. M., F. Kramarz, and D. N. Margolis (1999). High wage workers and high wage firms. *Econometrica 67*, 251–333.

Abowd, J. M., K. L. McKinney, and I. M. Schmutte (2019). Modeling endogenous mobility in earnings determination. *Journal of Business & Economic Statistics 37*, 405–418.

Allman, E. S., C. Matias, and J. A. Rhodes (2009). Identifiability of parameters in latent structure models with many observed variables. *Annals of Statistics 37*, 3099–3132.

Andrews, M. J., L. Gill, T. Schank, and R. Upward (2008). High wage workers and low wage firms: Negative assortative matching or limited mobility bias? *Journal of the Royal Statistical Society, Series A 171*, 673–697.

Azkarate-Askasua, M. and M. Zerecero (2024). No more limited mobility bias: Exploring the heterogeneity of labor markets. Mimeo.

Babet, D., O. Godechot, and M. G. Palladino (2025). In the land of AKM: Explaining the dynamics of wage inequality in France. Banque de France Working Paper Series 987.

Beffy, M., T. Kamionka, F. Kramarz, and C. Robert (2003). Job mobility and wages with worker and firm heterogeneity. Mimeo.

Bonhomme, S., K. Jochmans, and J.-M. Robin (2016a). Estimating multivariate latent-structure models. *Annals of Statistics 44*, 540–563.

Bonhomme, S., K. Jochmans, and J.-M. Robin (2016b). Nonparametric estimation of finite mixtures from repeated measurements. *Journal of the Royal Statistical Society, Series B 78*, 211–229.

Bonhomme, S., T. Lamadon, and E. Manresa (2019a). A distributional framework for matched employer-employee data. *Econometrica 87*, 699–738.

Bonhomme, S., T. Lamadon, and E. Manresa (2019b). Supplement to "a distributional framework for matched employer-employee data".

Burdett, K. and D. T. Mortensen (1998). Wage differentials, employer size, and unemployment. *International Economic Review 39*, 257–273.

Chetty, R., J. N. Friedman, and J. E. Rockoff (2014). Measuring the impacts of teachers I: Evaluating bias in teacher value-added estimates. *American Economic Review 104*, 2593–2632.

Delacroix, A. and S. Shi (2006). Directed search on the job and the wage ladder. *International*

Economic Review 47, 651–699.

Eeckhout, J. and P. Kircher (2011). Identifying sorting in theory. *Review of Economic Studies 78*, 872–906.

Hagedorn, M., T. H. Law, and I. Manovskii (2017). Identifying equilibrium models of labor market sorting. *Econometrica 85*, 29–65.

He, J. and J.-M. Robin (2025). Ridge estimation of high dimensional two-way fixed effect regression. Mimeo.

Higgins, A. and K. Jochmans (2023). Identification of mixtures of dynamic discrete choices. *Journal of Econometrics 237*, 105462.

Hu, Y. and M. Shum (2012). Nonparametric identification of dynamic models with unobserved state variables. *Journal of Econometrics 171*, 32–44.

Jochmans, K. and M. Weidner (2019). Fixed-effect regressions on network data. *Econometrica 87*, 1543–1560.

Kasahara, H. and K. Shimotsu (2014). Nonparametric identification and estimation of the number of components in multivariate mixtures. *Journal of the Royal Statistical Society, Series B 76*, 97–111.

Kline, P., R. Saggio, and M. Sølvsten (2020). Leave-out estimation of variance components. *Econometrica 88*, 1859–1898.

Kwon, C. and E. Mbakop (2021). Estimation of the number of components of non-parametric multivariate finite mixture models. *Annals of Statistics 49*, 2178–2205.

Lamadon, T., J. Lise, C. Meghir, and J.-M. Robin (2025). Labor market matching, wages, and amenities. Mimeo.

Lentz, R., S. Piyapromdee, and J.-M. Robin (2023). The anatomy of sorting - Evidence from Danish data. *Econometrica 91*, 2409–2455.

McCaffrey, D. F., J. R. Lockwood, D. Koretz, T. A. Louis, and L. Hamilton (2004). Models for value-added modeling of teacher effects. *Journal of Educational and Behavioral Statistics 29*, 67–101.

Schennach, S. M. (2020). Mismeasured and unobserved variables. In S. Durlauf, L. P. Hansen, J. J. Heckman, and R. Matzkin (Eds.), *Handbook of Econometrics*, Volume 7A, pp. 487–565. Elsevier.

Shimer, R. and L. Smith (2000). Assortative matching and search. *Econometrica 68*, 343–369.